

Unit – V

1. Four masses m_1, m_2, m_3 and m_4 are 200 kg, 300 kg, 240 kg and 260 kg respectively. The corresponding radii of rotation are 0.2 m, 0.15 m, 0.25 m and 0.3 m respectively and the angles between successive masses are $45^\circ, 75^\circ$ and 135° . Find the position and magnitude of the balance mass required, if its radius of rotation is 0.2 m.

Given:

$$m_1 = 200 \text{ kg}$$

$$m_2 = 300 \text{ kg}$$

$$m_3 = 240 \text{ kg}$$

$$m_4 = 260 \text{ kg}$$

$$r_1 = 0.2 \text{ m}$$

$$r_2 = 0.15 \text{ m}$$

$$r_3 = 0.25 \text{ m}$$

$$r_4 = 0.3 \text{ m}$$

$$\theta_1 = 0^\circ$$

$$\theta_2 = 45^\circ$$

$$\theta_3 = 45^\circ + 75^\circ = 120^\circ$$

$$\theta_4 = 45^\circ + 75^\circ + 135^\circ = 255^\circ$$

$$r = 0.2 \text{ m}$$

To Find:

$$\theta = ?$$

$$m = ?$$

Solution:

Let, m = Balancing mass, and

θ = the angle which the balancing mass makes with m_1 .

Since the magnitude of centrifugal forces are proportional to the product of each mass and its radius, therefore

$$m_1 \cdot r_1 = 200 \times 0.2 = 40 \text{ kg-m}$$

$$m_2 \cdot r_2 = 300 \times 0.15 = 45 \text{ kg-m}$$

$$m_3 \cdot r_3 = 240 \times 0.25 = 60 \text{ kg-m}$$

$$m_4 \cdot r_4 = 260 \times 0.3 = 78 \text{ kg-m}$$

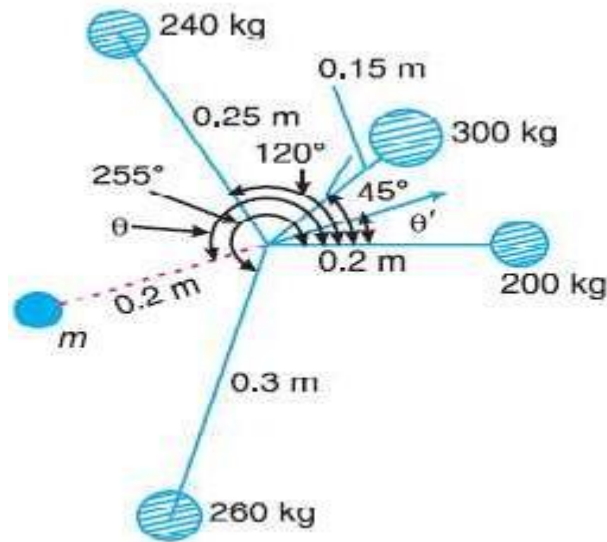
The problem may, now, be solved either analytically or graphically. But we shall solve the problem by both the methods one by one.

1. Analytical method

The space diagram is shown in Fig. 21.5.

Resolving $m_1 \cdot r_1, m_2 \cdot r_2, m_3 \cdot r_3$ and $m_4 \cdot r_4$ horizontally,

$$\begin{aligned}\Sigma H &= m_1 \cdot r_1 \cos \theta_1 + m_2 \cdot r_2 \cos \theta_2 + m_3 \cdot r_3 \cos \theta_3 + m_4 \cdot r_4 \cos \theta_4 \\ &= 40 \cos 0^\circ + 45 \cos 45^\circ + 60 \cos 120^\circ + 78 \cos 255^\circ \\ &= 40 + 31.8 - 30 - 20.2 = 21.6 \text{ kg-m}\end{aligned}$$



Now resolving vertically,

$$\begin{aligned}\Sigma V &= m_1 \cdot r_1 \sin \theta_1 + m_2 \cdot r_2 \sin \theta_2 + m_3 \cdot r_3 \sin \theta_3 + m_4 \cdot r_4 \sin \theta_4 \\ &= 40 \sin 0^\circ + 45 \sin 45^\circ + 60 \sin 120^\circ + 78 \sin 255^\circ \\ &= 0 + 31.8 + 52 - 75.3 = 8.5 \text{ kg-m}\end{aligned}$$

$$\therefore \text{Resultant, } R = \sqrt{(\Sigma H)^2 + (\Sigma V)^2} = (21.6)^2 + (8.5)^2 = 23.2 \text{ kg-m}$$

We know that

$$m \cdot r = R = 23.2 \text{ or}$$

$$m = 23.2 / r = 23.2 / 0.2 = \mathbf{116 \text{ kg Ans.}}$$

$$\text{And } \tan \theta' = \Sigma V / \Sigma H = 8.5 / 21.6 = 0.3935 \text{ or } \theta' = 21.48^\circ$$

Since θ' is the angle of the resultant R from the horizontal mass of 200 kg, therefore the angle of the balancing mass from the horizontal mass of 200 kg,

$$\theta = 180^\circ + 21.48^\circ = \mathbf{201.48^\circ \text{ Ans.}}$$

2. A shaft carries four masses A, B, C and D of magnitude 200 kg, 300 kg, 400 kg and 200 kg respectively and revolving at radii 80 mm, 70 mm, 60 mm and 80 mm in planes measured from A at 300 mm, 400 mm and 700 mm. The angles between the cranks measured anticlockwise are A to B 45° , B to C 70° and C to D 120° . The balancing masses are to be placed in planes X and Y. The distance between the planes A and X is 100 mm, between X and Y is 400 mm and between Y and D is 200 mm. If the balancing masses revolve at a radius of 100 mm, find their magnitudes and angular positions.

Given:

$$m_A = 200 \text{ kg}$$

$$m_B = 300 \text{ kg}$$

$$m_C = 400 \text{ kg}$$

$$m_D = 200 \text{ kg}$$

$$r_A = 80 \text{ mm} = 0.08 \text{ m}$$

$$r_B = 70 \text{ mm} = 0.07 \text{ m}$$

$$r_C = 60 \text{ mm} = 0.06 \text{ m}$$

$$r_D = 80 \text{ mm} = 0.08 \text{ m}$$

$$r_X = r_Y = 100 \text{ mm} = 0.1 \text{ m}$$

To Find:

$\theta_x = ?$

$\theta_y = ?$

$m_x = ?$

$m_y = ?$

Solution:

Let, m_X = Balancing mass placed in plane X, and

m_Y = Balancing mass placed in plane Y.

The position of planes and angular position of the masses (assuming the mass A as horizontal) are shown in Fig. 21.8 (a) and (b) respectively.

Assume the plane X as the reference plane (R.P.). The distances of the planes to the right of plane X are taken as +ve while the distances of the planes to the left of plane X are taken as ve.

The data may be tabulated as shown in Table.

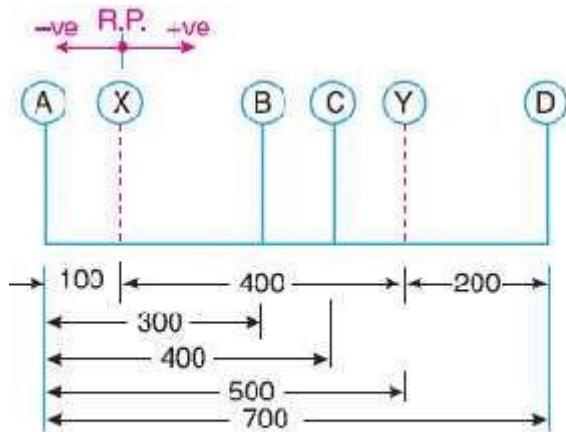
Plane (1)	Mass (m) kg (2)	Radius (r) m (3)	Cent. force + ω^2 (m.r) kg-m (4)	Distance from Plane X (l) m (5)	Couple + ω^2 (m.r.l) kg-m ² (6)
A	200	0.08	16	-0.1	-1.6
X(R.P.)	m_X	0.1	$0.1 m_X$	0	0
B	300	0.07	21	0.2	4.2
C	400	0.06	24	0.3	7.2
Y	m_Y	0.1	$0.1 m_Y$	0.4	$0.04 m_Y$
D	200	0.08	16	0.6	9.6

The balancing masses m_X and m_Y and their angular positions may be determined graphically as discussed below:

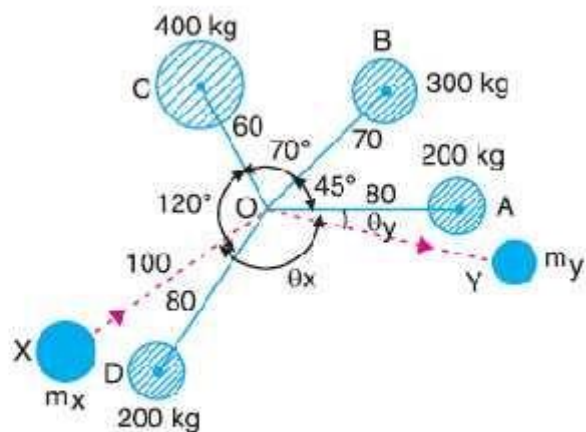
- First of all, draw the couple polygon from the data given in Table (column 6) as shown in Fig. 21.8 (c) to some suitable scale. The vector $d'o'$ represents the balanced couple. Since the balanced couple is proportional to $0.04 m_Y$, therefore by measurement,

$$0.004 m_Y = \text{vector } d'o' = 7.3 \text{ kg} - \text{m}^2;$$

$$m_Y = 182.5 \text{ kg}$$

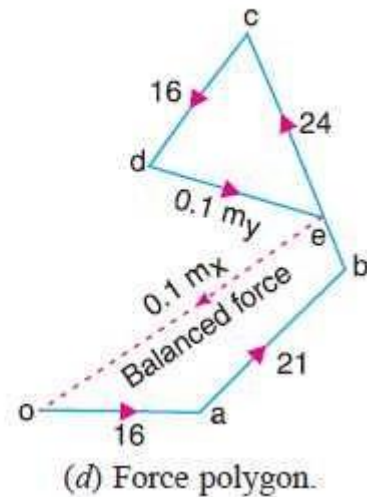
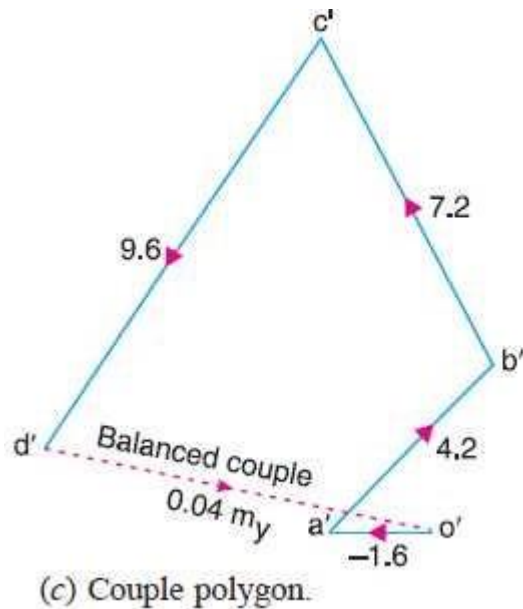


(a) Position of planes.



(b) Angular position of masses.

All dimensions in mm.



2. By measurement, the angular position of m_y is $\theta_y = 12^\circ$
3. Now draw the force polygon from the data given in Table (column 4) as shown in Fig.(d). The vector eo represents the balanced force. Since the balanced force is proportional to $0.1 m_x$, therefore by measurement,
 $0.1m_x = \text{vector } eo = 35.5 \text{ kg-m}$ or $m_x = 355 \text{ kg Ans.}$
4. By measurement, the angular position of m_x is $\theta_x = 145^\circ$

3. Four masses A, B, C and D as shown below are to be completely balanced.

	A	B	C	D
Mass (kg)	-	30	50	40
Radius(mm)	180	240	120	150

The planes containing masses B and C are 300 mm apart. The angle between planes containing B and C is 90° . B and C make angles of 210° and 120° respectively with D in the same sense. Find

1. The magnitude and the angular position of mass A; and
2. The position of planes A and D.

Given:

$$r_A = 180 \text{ mm} = 0.18 \text{ m}$$

$$m_B = 30 \text{ kg}$$

$$r_B = 240 \text{ mm} = 0.24 \text{ m}$$

$$m_C = 50 \text{ kg}$$

$$r_C = 120 \text{ mm} = 0.12 \text{ m}$$

$$m_D = 40 \text{ kg}$$

$$r_D = 150 \text{ mm} = 0.15 \text{ m}$$

$$\angle BOC = 90^\circ$$

$$\angle BOD = 210^\circ$$

$$\angle COD = 120^\circ$$

To Find:

$$m_A = ?$$

$$\angle AOB = ?$$

Positions of plane A and D = ?

Solution:

1. The magnitude and the angular position of mass A

Let m_A = Magnitude of Mass A,

x = Distance between the planes B and D, and

y = Distance between the planes A and B.

The position of the planes and the angular position of the masses are shown in Fig (a) and (b) respectively.

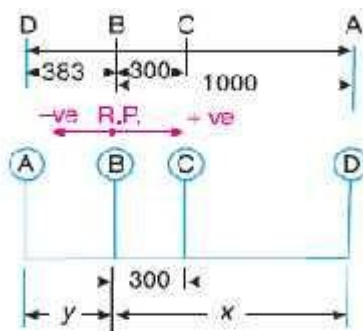
Assuming the plane B as the reference plane (R.P.) and the mass B (m_B) along the horizontal line as shown in Fig (b), the data may be tabulated as below:

Plane (1)	Mass (m) kg (2)	Radius (r) m (3)	Cent. force = ω^2 (m.r) kg-m (4)	Distance from plane B (l) m (5)	Couple = ω^2 (m.r.l) kg-m ² (6)
A	m_A	0.18	$0.08 m_A$	y	$0.18 m_A y$
B (R.P.)	30	0.24	7.2	0	0
C	50	0.12	6	0.3	1.8
D	40	0.15	6	x	$6x$

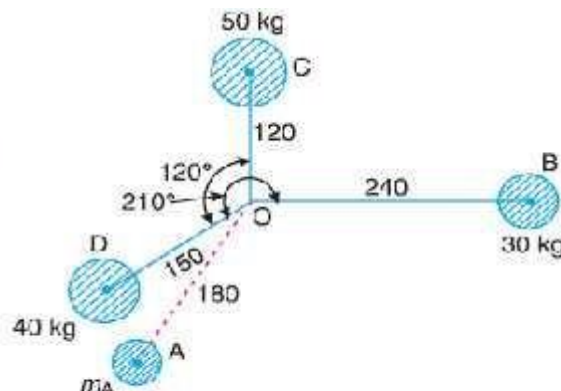
The magnitude and angular position of mass A may be determined by drawing the force polygon from the data given in Table (Column 4), as shown in Fig (c), to some suitable scale. Since the masses are to be completely balanced, therefore the force polygon must be a closed figure. The closing side (i.e. vector do) is proportional to $0.18 m_A$. By measurement,

$$0.18 m_A = \text{Vector } do = 3.6 \text{ kg-m or } m_A = 20 \text{ kg Ans.}$$

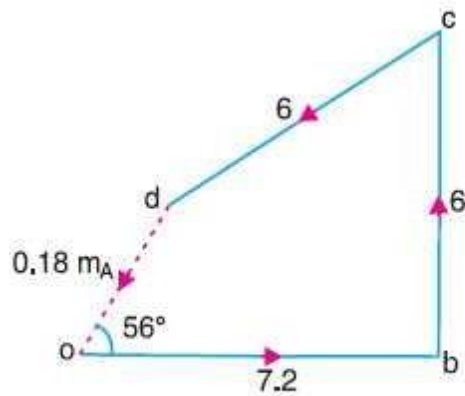
In order to find the angular position of mass A, draw OA in Fig. 21.9 (b) parallel to vector do . By measurement, we find that the angular position of mass A from mass B in the anticlockwise direction is $\angle AOB = 236^\circ$ Ans.



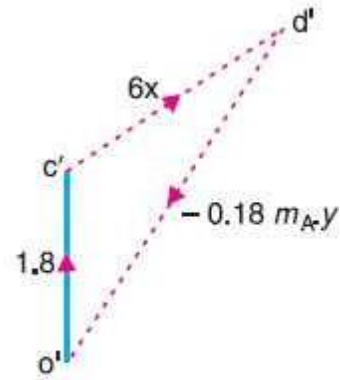
(a) Position of planes.



(b) Angular position of masses.



(c) Force polygon.



(d) Couple polygon.

2. Position of planes A and D

The position of planes A and D may be obtained by drawing the couple polygon, as shown in Fig(d), from the data given in Table 21.3 (column 6). The couple polygon is drawn as discussed below :

1. Draw vector $c'd'$ parallel to OC and equal to 1.8 kg-m^2 , to some suitable scale.
2. From point's c' and d' , draw lines parallel to OD and OA respectively, such that they intersect at point d' . By measurement, we find that

$$6x = \text{vector } c'd' = 2.3 \text{ kg-m}^2 \text{ or } x = 0.383 \text{ m}$$

We see from the couple polygon that the direction of vector $c'd'$ is opposite to the direction of mass D. Therefore the plane of mass D is 0.383 m or 383 mm towards left of plane B and not towards right of plane B as already assumed. **Ans**

Again by measurement from couple polygon,

$$-0.18 m_A y = \text{vector } o'd' = 3.6 \text{ kg-m}^2$$

$$-0.18 \times 20 y = 3.6 \text{ or } y = -1 \text{ m}$$

The negative sign indicates that the plane A is not towards left of B as assumed but it is 1 m or 1000 mm towards right of plane B.

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4. A, B, C and D are four masses carried by a rotating shaft at radii 100, 125, 200 and 150 mm respectively. The planes in which the masses revolve are spaced 600 mm apart and the mass of B, C and D are 10 kg, 5 kg, and 4 kg respectively. Find the required mass A and the relative angular settings of the four masses so that the shaft shall be in complete balance.

Given:

$$r_A = 100 \text{ mm} = 0.1 \text{ m}$$

$$r_B = 125 \text{ mm} = 0.125 \text{ m}$$

$$r_C = 200 \text{ mm} = 0.2 \text{ m}$$

$$r_D = 150 \text{ mm} = 0.15 \text{ m}$$

$$m_B = 10 \text{ kg} ; m_C = 5 \text{ kg} ; m_D = 4 \text{ kg}$$

To Find:

$\angle BOC = ?$

$\angle BOD = ?$

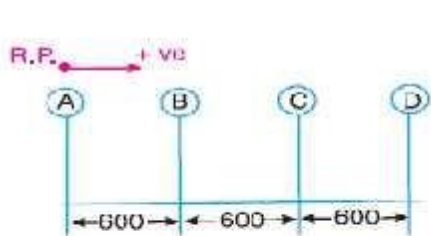
$\angle BOA = ?$

$m_A = ?$

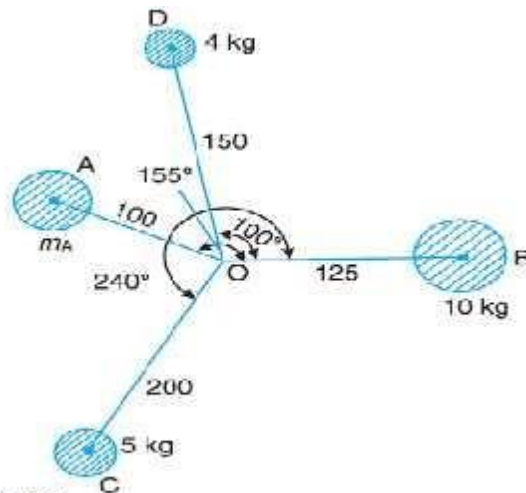
Solution:

The position of planes is shown in Fig (a). Assuming the plane of mass A as the reference plane ($R.P.$), the data may be tabulated as below:

Plane (1)	Mass (m) kg (2)	Radius (r) m (3)	Cent. Force + ω^2 ($m.r$) kg-m (4)	Distance from plane A (l)m (5)	Couple + ω^2 ($m.r.l$) kg-m ² (6)
A(R.P.)	m_A	0.1	$0.1 m_A$	0	0
B	10	0.125	1.25	0.6	0.75
C	5	0.2	1	1.2	1.2
D	4	0.15	0.6	1.8	1.08



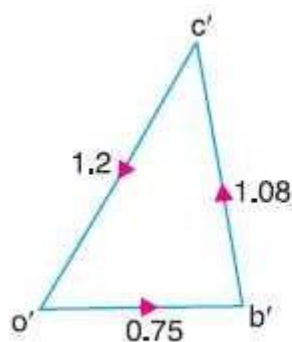
(a) Position of planes.



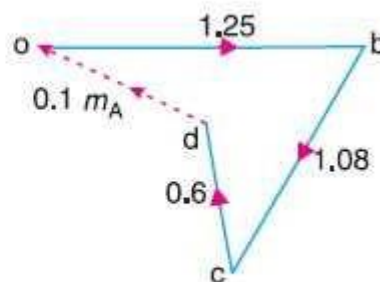
All dimensions in mm

(b) Angular position of masses.

First of all, the angular setting of masses C and D is obtained by drawing the couple polygon from the data given in Table (column 6). Assume the position of mass B in the horizontal direction OB as shown in Fig (b). Now the couple polygon as shown in Fig (c) is drawn as discussed below:



(c) Couple polygon.



(d) Force polygon.

1. Draw vector $o'b'$ in the horizontal direction (i.e. parallel to OB) and equal to 0.75 kg-m², to some suitable scale.
2. From points o' and b' , draw vectors $o'c'$ and $b'c'$ equal to 1.2 kg-m² and 1.08 kg-m² respectively. These vectors intersect at c' .

3. Now in Fig. 21.10 (b), draw OC parallel to vector $d'c'$ and OD parallel to vector $b'c'$.

By measurement, we find that the angular setting of mass C from mass B in the anticlockwise direction, *i.e.*

$$\angle BOC = 240^\circ \text{ Ans.}$$

and angular setting of mass D from mass B in the anticlockwise direction, *i.e.*

$$\angle BOD = 100^\circ \text{ Ans.}$$

In order to find the required mass A (m_A) and its angular setting, draw the force polygon to some suitable scale, as shown in Fig. 21.10 (d), from the data given in Table (column 4).

Since the closing side of the force polygon (vector do) is proportional to $0.1 m_A$, therefore by measurement,

$$0.1 m_A = 0.7 \text{ kg-m}^2 \text{ or } m_A = 7 \text{ kg Ans.}$$

Now draw OA in Fig. 21.10 (b), parallel to vector do . By measurement, we find that the angular setting of mass A from mass B in the anticlockwise direction, *i.e.*

$$\angle BOA = 155^\circ \text{ Ans.}$$

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5. A shaft carries four masses in parallel planes A, B, C and D in this order along its length. The masses at B and C are 18 kg and 12.5 kg respectively, and each has an eccentricity of 60 mm . The masses at A and D have an eccentricity of 80 mm . The angle between the masses at B and C is 100° and that between the masses at B and A is 190° , both being measured in the same direction. The axial distance between the planes A and B is 100 mm and that between B and C is 200 mm . If the shaft is in complete dynamic balance, determine:

1. The magnitude of the masses at A and D ;
2. The distance between planes A and D
3. The angular position of the mass at D .

Given

$$m_B = 18 \text{ kg}$$

$$m_C = 12.5 \text{ kg}$$

$$r_B = r_C = 60 \text{ mm} = 0.06 \text{ m}$$

$$r_A = r_D = 80 \text{ mm} = 0.08 \text{ m}$$

$$\angle BOC = 100^\circ; \angle BOA = 190^\circ$$

To Find:

$$m_A = ?$$

$$m_D = ?$$

$$X = ?$$

$$\angle BOD = ?$$

Solution:

1. Magnitude of the masses at A and D

Let, m_A = Mass at A ,

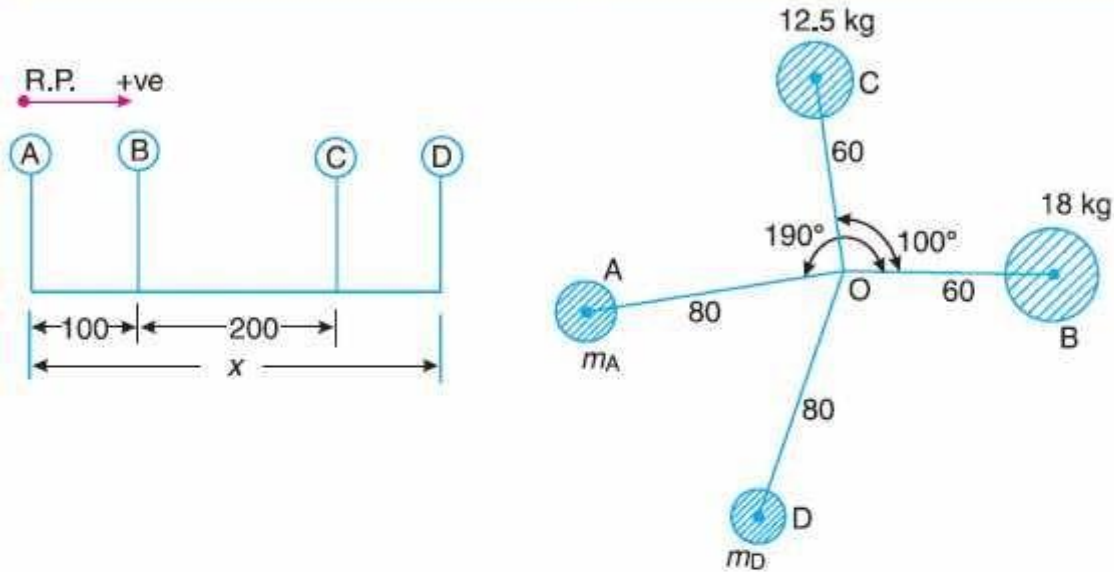
$$m_D = \text{Mass at } D, \text{ and}$$

$$x = \text{Distance between planes } A \text{ and } D.$$

The position of the planes and angular position of the masses is shown in Fig (a) and (b) respectively. The position of mass *B* is assumed in the horizontal direction, i.e. along *OB*.

Taking the plane of mass *A* as the reference plane, the data may be tabulated as below:

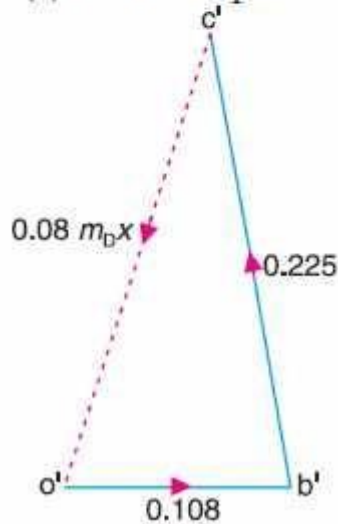
Plane (1)	Mass (<i>m</i>) kg (2)	Eccentricity (<i>r</i>) m (3)	Cent. force $\div \omega^2$ (<i>m.r</i>) kg-m (4)	Distance from plane <i>A</i> (<i>l</i>)m (5)	Couple $\div \omega^2$ (<i>m.r.l</i>) kg-m ² (6)
<i>A</i> (R.P.)	m_A	0.08	$0.08 m_A$	0	0
<i>B</i>	18	0.06	1.08	0.1	0.108
<i>C</i>	12.5	0.06	0.75	0.3	0.225
<i>D</i>	m_D	0.08	$0.08 m_D$	<i>x</i>	$0.08 m_D \cdot x$



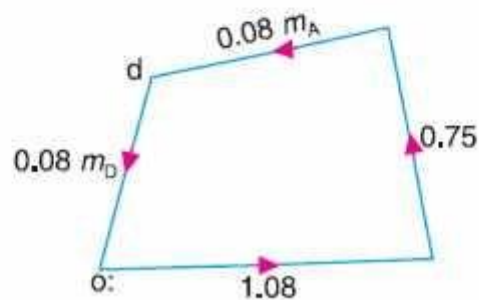
All dimensions in mm.

(a) Position of planes.

(b) Angular position of masses.



(c) Couple polygon.



(d) Force polygon.

First of all, the direction of mass *D* is fixed by drawing the couple polygon to some suitable scale, as shown in Fig (c), from the data given in Table (column 6). The closing side of the couple polygon (vector *c' o'*) is proportional to $0.08 m_D \cdot x$. By measurement, we find that

$$0.08 m_D \cdot x = \text{vector } c' o' = 0.235 \text{ kg-m}^2 \dots (i)$$

In Fig (b), draw OD parallel to vector $c'o'$ to fix the direction of mass D .

Now draw the force polygon, to some suitable scale, as shown in Fig (d), from the data given in Table 21.5 (column 4), as discussed below :

1. Draw vector ob parallel to OB and equal to 1.08 kg-m .
2. From point b , draw vector bc parallel to OC and equal to 0.75 kg-m .
3. For the shaft to be in complete dynamic balance, the force polygon must be a closed figure. Therefore from point c , draw vector cd parallel to OA and from point o draw vector od parallel to OD . The vectors cd and od intersect at d . Since the vector cd is proportional to 0.08 mA , therefore by measurement

$$0.08 \text{ mA} = \text{vector } cd = 0.77 \text{ kg-m or } \mathbf{mA = 9.625 \text{ kg Ans.}}$$

and vector do is proportional to 0.08 mD , therefore by measurement,

$$0.08 \text{ mD} = \text{vector } do = 0.65 \text{ kg-m or } \mathbf{mD = 8.125 \text{ kg Ans.}}$$

2. Distance between planes A and D

From equation (i),

$$0.08 \text{ mD} \cdot x = 0.235 \text{ kg-m}^2$$

$$0.08 \times 8.125 \times x = 0.235 \text{ kg-m}^2 \text{ or } 0.65 x = 0.235$$

$$\mathbf{x = 361.5 \text{ mm Ans.}}$$

3. Angular position of mass at D

By measurement from Fig. 21.11 (b), we find that the angular position of mass at D from mass B in the anticlockwise direction, i.e. $\angle BOD = 251^\circ \text{ Ans.}$

6. A shaft has three eccentrics, each 75 mm diameter and 25 mm thick, machined in one piece with the shaft. The central planes of the eccentric are 60 mm apart. The distance of the centres from the axis of rotation are 12 mm, 18 mm and 12 mm and their angular positions are 120° apart. The density of metal is 7000 kg/m^3 . Find the amount of out-of-balance force and couple at 600 r.p.m. If the shaft is balanced by adding two masses at a radius 75 mm and at distances of 100 mm from the central plane of the middle eccentric, find the amount of the masses and their angular positions.

Given:

$$D = 75 \text{ mm} = 0.075 \text{ m}$$

$$t = 25 \text{ mm} = 0.025 \text{ m}$$

$$r_A = 12 \text{ mm} = 0.012 \text{ m}$$

$$r_B = 18 \text{ mm} = 0.018 \text{ m}$$

$$r_C = 12 \text{ mm} = 0.012 \text{ m}$$

$$\rho = 7000 \text{ kg/m}^3$$

$$N = 600 \text{ r.p.m. or } \omega = 2\pi \times 600/60 = 62.84 \text{ rad/s}$$

$$r_L = r_M = 75 \text{ mm} = 0.075 \text{ m}$$

Solution:

We know that mass of each eccentric,

$$m_A = m_B = m_C = \text{Volume} \times \text{Density} = \frac{\pi}{4} \times D^2 \times t \times \rho$$
$$= 0.77 \text{ kg}$$

Let L and M be the planes at distances of 100 mm from the central plane of middle eccentric.

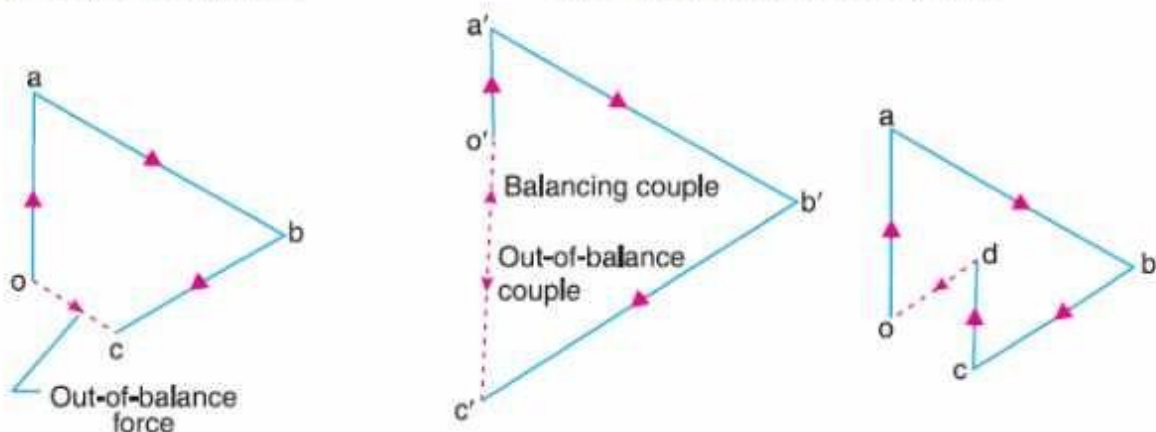
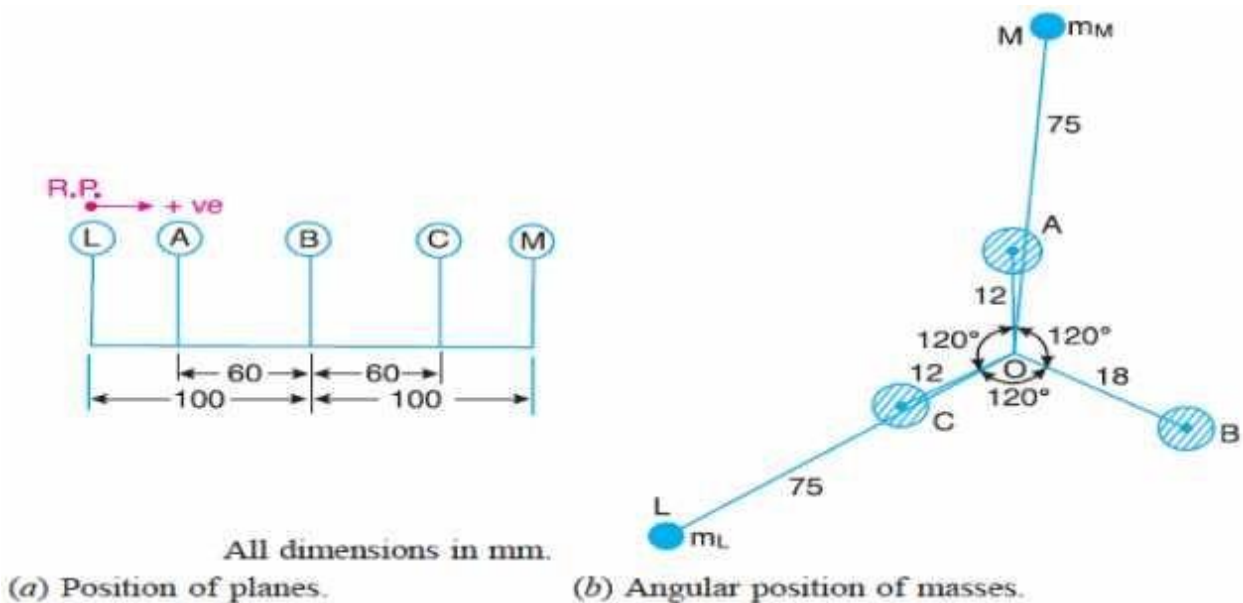
The position of the planes and the angular position of the three eccentrics is shown in

Fig (a) and (b) respectively, Assuming L as the reference plane and mass of the eccentric A in the vertical direction, the data may be tabulated as below:

Plane (1)	Mass (m) kg (2)	Radius (r) m (3)	Cent. force $= \omega^2$ (m.r) kg-m (4)	Distance from plane L.(l)m (5)	Couple $= \omega^2$ (m.r.l) kg-m ² (6)
L (R.P.)	m_L	0.075	$75 \times 10^{-3} m_L$	0	0
A	0.77	0.012	9.24×10^{-3}	0.04	0.3696×10^{-3}
B	0.77	0.018	13.86×10^{-3}	0.1	1.386×10^{-3}
C	0.77	0.012	9.24×10^{-3}	0.16	1.4784×10^{-3}
M	m_M	0.075	$75 \times 10^{-3} m_M$	0.20	$15 \times 10^{-3} m_M$

Out-of-balance force

The out-of-balance force is obtained by drawing the force polygon, as shown in Fig (c), from the data given in Table (column 4). The resultant oc represents the out-of-balance force.



$$oa = 9.24 \times 10^{-3}$$

$$ab = 13.86 \times 10^{-3}$$

$$bc = 9.24 \times 10^{-3}$$

(c) Force polygon.

$$o'a' = 0.3696 \times 10^{-3}$$

$$a'b' = 1.386 \times 10^{-3}$$

$$b'c' = 1.4784 \times 10^{-3}$$

(d) Couple polygon.

$$oa = 9.24 \times 10^{-3}$$

$$ab = 13.86 \times 10^{-3}$$

$$bc = 9.24 \times 10^{-3}$$

$$cd = 75 \times 10^{-3} m_M$$

(e) Force polygon.

Since the centrifugal force is proportional to the product of mass and radius (*i.e.* $m.r$), therefore by measurement.

$$\text{Out-of-balance force} = \text{vector } oc = 4.75 \times 10^{-3} \text{ kg-m}$$

$$= 4.75 \times 10^{-3} \times \omega^2 = 4.75 \times 10^{-3} (62.84)^2 = 18.76 \text{ N } \textbf{Ans.}$$

Out-of-balance couple

The out-of-balance couple is obtained by drawing the couple polygon from the data given in Table (column 6), as shown in Fig(d). The resultant $d'c'$ represents the out-of balance couple. Since the couple is proportional to the product of force and distance ($m.r.l$), therefore by measurement,

$$\text{Out-of-balance couple} = \text{vector } d'c' = 1.1 \times 10^{-3} \text{ kg-m}^2$$

$$= 1.1 \times 10^{-3} \times \omega^2 = 1.1 \times 10^{-3} (62.84)^2 = 4.34 \text{ N-m } \textbf{Ans.}$$

Amount of balancing masses and their angular positions

The vector $c'd'$ (in the direction from c' to d'), as shown in Fig (d) represents the balancing couple and is proportional to $15 \times 10^{-3} m_M$, *i.e.*

$$15 \times 10^{-3} m_M = \text{vector } c'd' = 1.1 \times 10^{-3} \text{ kg-m}^2$$

$$\text{or } m_M = 0.073 \text{ kg } \textbf{Ans.}$$

Draw OM in Fig. 21.12 (b) parallel to vector $c'd'$. By measurement, we find that the angular position of balancing mass (m_M) is 5° from mass A in the clockwise direction. **Ans.**

In order to find the balancing mass (m_L), a force polygon as shown in Fig (e) is drawn. The closing side of the polygon *i.e.* vector do (in the direction from d to o) represents the balancing force and is proportional to $75 \times 10^{-3} m_L$. By measurement, we find that

$$75 \times 10^{-3} m_L = \text{vector } do = 5.2 \times 10^{-3} \text{ kg-m}$$

$$\text{or } m_L = 0.0693 \text{ kg } \textbf{Ans.}$$

Draw OL in Fig (b), parallel to vector do . By measurement, we find that the angular position of mass (m_L) is 124° from mass A in the clockwise direction. **Ans.**

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7. A shaft is supported in bearings 1.8 m apart and projects 0.45 m beyond bearings at each end. The shaft carries three pulleys one at each end and one at the middle of its length. The mass of end pulleys is 48 kg and 20 kg and their centre of gravity are 15 mm and 12.5 mm respectively from the shaft axis. The centre pulley has a mass of 56 kg and its centre of gravity is 15 mm from the shaft axis. If the pulleys are arranged so as to give static balance, determine: **1.** relative angular positions of the pulleys, and **2.** dynamic forces produced on the bearings when the shaft rotates at 300 r.p.m.

Given:

$$\begin{aligned}
 m_A &= 48 \text{ kg;} \\
 m_C &= 20 \text{ kg;} \\
 r_A &= 15 \text{ mm} = 0.015 \text{ m;} \\
 r_C &= 12.5 \text{ mm} = 0.0125 \text{ m;} \\
 m_B &= 56 \text{ kg;} \\
 r_B &= 15 \text{ mm} = 0.015 \text{ m;} \\
 N &= 300 \text{ r.p.m. or} \\
 \omega &= 2\pi \times 300/60 = 31.42 \text{ rad/s}
 \end{aligned}$$

To Find:

Angle between pulleys B and A = ?
 Angle between pulleys A and C = ?
 Angle between pulleys C and B = ?
 Dynamic force at the bearing M = ?
 Dynamic force at the bearing L = ?

Solution:

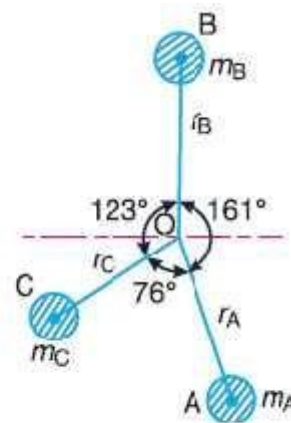
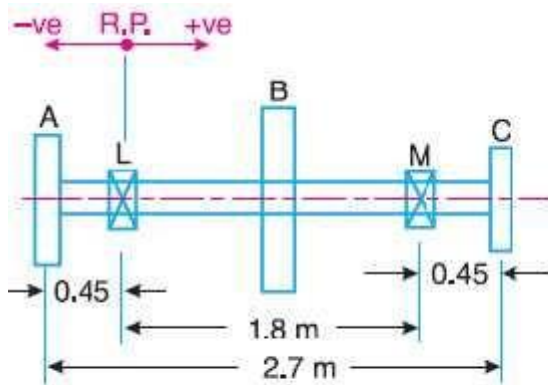
1. Relative angular position of the pulleys

The position of the shaft and pulleys is shown in Fig (a).

Let m_L and m_M = Mass at the bearings L and M , and

r_L and r_M = Radius of rotation of the masses at L and M respectively.

Assuming the plane of bearing L as reference plane, the data may be tabulated as below:



(a) Position of shaft and pulleys.

(b) Angular position of pulleys.

Plane (1)	Mass (m) kg (2)	Radius (r) m (3)	Cent. force $\times \omega^2$ (m.r) kg-m (4)	Distance from plane L (l) m (5)	Couple $= \omega^2$ (m.r.l) kg-m ² (6)
A	48	0.015	0.72	0.45	0.324
L(R.P)	m_L	r_L	$m_L \cdot r_L$	0	0
B	56	0.015	0.84	0.9	0.756
M	m_M	r_M	$m_M \cdot r_M$	1.8	$1.8 m_M \cdot r_M$
C	20	0.0125	0.25	2.25	0.5625

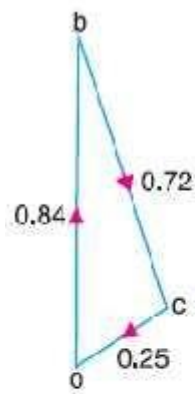
First of all, draw the force polygon to some suitable scale, as shown in Fig.(c), from the data given in Table (column 4).

By measurement, we find that

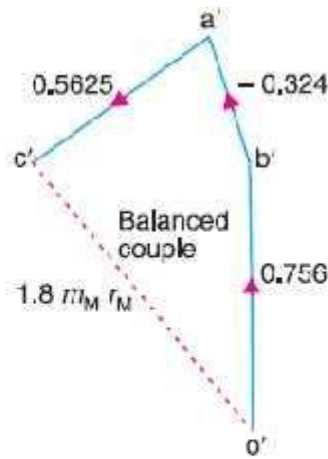
Angle between pulleys B and A = 161° **Ans.**

Angle between pulleys A and C = 76° **Ans.**

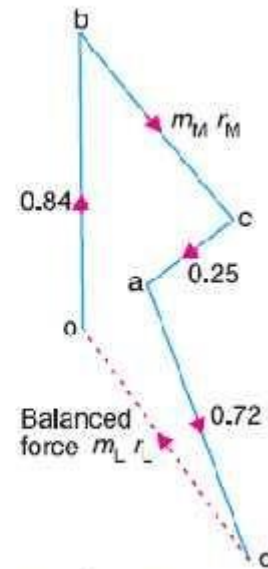
Angle between pulleys C and B = 123° **Ans.**



(c) Force polygon.



(d) Couple polygon.



(e) Force polygon.

2. Dynamic forces at the two bearings

In order to find the dynamic forces (or reactions) at the two bearings L and M , let us first calculate the values of $m_L.r_L$ and $m_M.r_M$ as discussed below:

1. Draw the couple polygon to some suitable scale, as shown in Fig (d), from the data given in Table (column 6). The closing side of the polygon (vector $c'o'$) represents the balanced couple and is proportional to $1.8m_M.r_M$. By measurement, we find that $1.8m_M.r_M = \text{vector } c'o' = 0.97 \text{ kg-m}^2$ or $m_M.r_M = 0.54 \text{ kg-m}$

$\therefore$ Dynamic force at the bearing M

$$m_M.r_M.\omega^2 = 0.54 (31.42)^2 = 533 \text{ N Ans.}$$

2. Now draw the force polygon, as shown in Fig (e), from the data given in Table (column 4) and taking $m_M.r_M = 0.54 \text{ kg-m}$. The closing side of the polygon (vector do) represents the balanced force and is proportional to $m_L.r_L$. By measurement, we find that

$$m_L.r_L = 0.54 \text{ kg-m}$$

$\therefore$ Dynamic force at the bearing L

$$m_L.r_L.\omega^2 = 0.54 (31.42)^2 = 533 \text{ N Ans.}$$